

LOT COMPUTER PAGER-0

Hardware program manual · plan stage · v0.1
LOT Systems Corporation · 2026-10-05
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01 — PLAN & ROADMAP

1. Product definition

Item	Spec
Form	40 × 40 mm square, 5 mm target thickness (see risk R1)
Body	Two-part 316L stainless steel (front bezel + back plate)
Back	Flat, polished stainless, no openings
Front	Small screen, camera, one button ("Copy")
Compute/radio	ESP32-S3 class SoC, Wi-Fi + BLE
Sensing	Weather: temperature / humidity / pressure (+ gas, AI-grade)
Power	Thin Li-Po, wireless (Qi) charging
Link	HTTPS to lot-systems.com via LOT API connector
Run	100 units, PCB + assembly by PCBWay

Behaviour:

1. Site's AI engine decides a nudge is due ("Coffee time!") and queues it for the device.
2. Device fetches it, shows it on screen, buzz/chirp (pager-like).
3. Button "Copy" acknowledges; device POSTs a log event; it appears in the Log tab.
4. Device periodically posts weather readings; camera captures on demand only.

2. Brief-to-plan traceability (all 19 items)

#	Brief item	Where handled
1	PCBWay	02 (PCB/assembly lines), 05 §4
2	Pager-like AI notification	04 §2
3	2-part stainless body	05 §1
4	4×4 cm × 5 mm flat square	01 R1, 05 §1
5	Camera	02, 03 §5, 05 §6
6	LOT API connector	04
7	PDF manuals	manuals/, 03 §7
8	Compress info each session	README digest, sessions/
9	Firmware documents	03
10	Software to connect with firmware	04 §4
11	Separate documents	this set
12	Charger	05 §3
13	100-unit run	02 §3, 01 §5
14	Weather sensor	02, 03
15	AI-grade off-the-shelf sensors	02 (BME688 + BHI260AP optional)
16	"Copy" button → Log tab	04 §3
17	Polished stainless side	05 §1
18	Camera + screen + button side	05 §1
19	Wireless charger	05 §3

3. Risk register

ID	Risk	Severity	Mitigation
R1	5 mm total thickness: shell walls (2×0.6) + PCB (0.8) + display (≥1.2) + cell (≥2.0) + camera (≥2.5 Z) cannot be stacked.	HIGH	Side-by-side layout (camera/button beside screen), flex PCB, 2 mm pouch cell (~60–80 mAh). EVT at 7–8 mm to prove function, squeeze to 5 mm in DVT. If it fails, ship 6.5 mm and say so.
R2	Stainless steel attenuates Qi (eddy currents).	HIGH	Charge through the front (glass) face, or non-metal inlay in back plate. Test both in EVT. See 05 §3.

R3	Metal case detunes the Wi-Fi/BLE antenna.	HIGH	Plastic/ceramic antenna window or gap between the two steel parts. Antenna keep-out; test in EVT.
R4	Camera privacy (always-near-person device).	MED	No always-on capture. Capture only on button, hardware LED lit, no audio. Document in manual.
R5	Tiny battery (~70 mAh) gives short runtime.	MED	Deep sleep + wake-on-timer polling (every 60–300 s); screen on only for notifications. Target ≥3 days.
R6	Lithium cell shipping and certification (UN38.3, FCC, CE, Qi).	MED	Pre-certified radio module; budget for test lab; 05 \$5.
R7	Stainless machining cost at 100 units.	MED	Quote CNC vs MIM; polish as separate op. 05 \$1.
R8	Brand sites unreadable this session.	LOW	Re-run when egress allows, or Vadik pastes brand rules.

4. Roadmap (calendar assumes start Oct 2026)

Phase	Window	Exit gate
P0 Plan (this doc)	Oct 2026	Vadik approves BOM direction + decisions §6
P1 Software first	Oct–Nov 2026	/api/device/* endpoints live; simulator device passes round trip (notify → show → Copy → Log tab)
P2 EVT dev boards	Nov–Dec 2026	ESP32-S3 dev kit + display + BME688 + camera on breadboard, talking to staging
P3 EVT PCB v1 (PCBWay proto, 5–10 pcs)	Dec 2026–Jan 2027	Board boots, Wi-Fi OK, Copy → Log works, charging works on a bare coil
P4 Enclosure prototypes (3D-printed steel / CNC 3–5 pcs)	Jan–Feb 2027	RF + Qi measured inside steel; thickness verdict
P5 DVT (20 units)	Mar–Apr 2027	Pre-compliance pass, battery life measured, manuals v1
P6 Compliance	Apr–Jun 2027	FCC/CE/Qi/UN38.3 reports
P7 PVT / 100-unit run	Jun–Aug 2027	100 units shipped, yield >90%

Dates are planning estimates, not commitments.

5. 100-unit run plan

- Buy parts for 120 (20% attrition for rework, steel scrap, DOA).
- PCBWay: PCBA turnkey, 100 + 5 spares; request X-ray on any BGA/QFN with thermal pad.
- Steel: 110 shells (2 parts each) to allow polishing rejects.
- Per-unit flash + provisioning: device ID + token via the host tool (04 §4); log in a CSV.
- Test fixture: pogo-pin jig, runs the self-test (03 §6), prints pass/fail.
- Estimated unit cost at 100 qty: see 02 §3 (roughly \$95–\$170/unit, dominated by steel + assembly).

6. Decisions needed from Vadik

1. Accept EVT at 7–8 mm, with 5 mm as the DVT goal? (Recommended: yes.)
2. Charging from the front/screen side vs. a non-metal inlay on the polished back? (Recommended: test both; front first.)
3. Camera: still image on button, or also a periodic image to the site? (Recommended: on-demand only.)
4. Screen: 1.14" colour IPS (recommended) vs 0.96" OLED vs 1.54" e-paper.
5. Provide brand colours/logo if different from LOT-STYLE-GUIDE (brand site was unreachable).

02 — COMPONENTS BUYING LIST

IMPORTANT: Prices are order-of-magnitude estimates from general knowledge.

Links go to vendor sites/search pages; none were opened or price-checked this session (no live parts lookup available). Verify part number, stock, thickness, and datasheet before ordering. Parts marked VERIFY are the most uncertain.

1. EVT prototype buy (dev stage, ~\$150–300)

Qty	Item	Why	Where
2	ESP32-S3 dev board with PSRAM (e.g. Seeed XIAO ESP32S3 Sense, includes OV2640 camera)	Fastest path to Wi-Fi + camera + firmware work	https://www.seeedstudio.com/
2	1.14" 240×135 IPS (ST7789)	Screen trial	https://www.adafruit.com/ , https://www.aliexpress.com/
2	BME688 breakout	Weather + gas	https://www.adafruit.com/ , https://www.sparkfun.com/
2	Qi receiver coil + RX board (5 V, thin)	Charging-through-steel test	https://www.sparkfun.com/ , https://www.digikey.com/
1	Qi transmitter pad	Test source	any WPC-certified pad
5	302025-class Li-Po (3 mm, ~120 mAh) and 2 mm pouch cells	Battery thickness study	https://www.digikey.com/ , https://www.adafruit.com/
1	316L stainless sheet samples 0.5 / 0.8 mm	RF + Qi attenuation test	https://www.mcmaster.com/
1	Coin-size piezo buzzer / LRA haptic	Pager alert	https://www.digikey.com/

2. Production BOM (per unit, target)

Block	Part (candidate)	Notes	Est. \$ @100	Link
MCU+radio	Espressif ESP32-S3-MINI-1 (pick variant with PSRAM; VERIFY)	Pre-certified module; 2.4 GHz Wi-Fi/BLE	4	https://www.espressif.com/en/products/modules
Camera	Ultra-thin 2 MP FPC module, Z ≤ 3 mm (OV2640 or GC2145 class; VERIFY thickness)	Biggest height risk	6–10	https://www.alibaba.com/ (search "ultra thin camera module DVP")
Display	1.14" IPS 240×135 ST7789, FPC, ≤1.5 mm	Alt: 0.96" OLED	4–7	https://www.alibaba.com/
Weather/gas	Bosch BME688 (2.5×2.5×0.93 mm)	"AI-grade" gas sensor w/ BSEC	9–12	https://www.digikey.com/
Motion (opt.)	Bosch BHI260AP smart IMU w/ on-chip AI	Tap/wake gestures	8	https://www.digikey.com/
Qi RX	TI BQ51003-class Qi receiver IC (VERIFY)	5 V out	3	https://www.ti.com/
Charger/PMIC	TI BQ25100-class Li-Ion charger + 3.3 V buck/LDO		2	https://www.ti.com/
Fuel gauge (opt.)	MAX17048	% battery	2	https://www.analog.com/
Qi coil	Thin RX coil + ferrite, ~Ø20 mm, ≤0.6 mm		2–3	https://www.digikey.com/
Battery	Li-Po 2–3 mm thick, 60–120 mAh, with PCM	Custom size likely	3–5	https://www.alibaba.com/
Button	Ultra-low-profile tact (≤0.6 mm) or capacitive/domed		0.5	https://www.digikey.com/
Haptic/buzzer	Thin piezo or LRA	Pager alert	1–2	https://www.digikey.com/
Status LED	Camera-active LED	Privacy	0.1	https://www.digikey.com/

PCB	4-layer, 0.6–0.8 mm, rigid-flex or flex-rigid (VERIFY need)	PCBWay	8–15	https://www.pcbway.com/
Assembly	PCBWay turnkey PCBA		10–20	https://www.pcbway.com/pcb-assembly.html
Steel body	2-part 316L CNC/MIM, polished back		30–70	https://www.pcbway.com/cnc-machining.html (also Xometry, JLCNC)
Window/inlay	Glass cover + antenna window		3–8	custom
Misc	adhesive, gasket, screws, label, packaging		5–10	—

3. 100-unit run estimate

Line	Est. total
Parts + PCB + assembly (120 sets)	\$2,500–\$4,500
Steel bodies (110 sets, incl. polish)	\$4,000–\$8,000
NRE: enclosure tooling/fixtures, test jig	\$1,500–\$4,000
Compliance pre-scan (FCC/CE/Qi/UN38.3)	\$5,000–\$15,000 (not per-unit; may be deferred)
Packaging + charger pads (100 pcs, ~\$6 each)	\$1,000–\$1,500
Total (excl. compliance)	~\$9,000–\$18,000 → ~\$95–\$170/unit

Add compliance if the units will be sold; skip for internal/prototype-only run.

4. Chargers

- Wireless: WPC Qi 5 W transmitter pad, flat, ~\$5–10 in 100 qty; or a custom LOT-branded stainless puck with a Qi TX module (later).
- Wired fallback: 4 pogo pads on the edge or USB-C is impossible at 5 mm, so use pogo pads on the front bezel for the factory flash/test jig only.

5. Where the order happens

1. PCBWay account → quote PCBA with this BOM (upload Gerbers + BOM + pick-and-place at P3).
2. Steel: request quotes from 3 shops (PCBWay CNC, Xometry, a local polisher).
3. Dev parts: Adafruit / Seeed / DigiKey carts (list in §1).

03 — FIRMWARE SPEC

1. Stack

- MCU: ESP32-S3, ESP-IDF 5.x (C/C++), FreeRTOS. Arduino-ESP32 acceptable for EVT only.
- OTA: dual-partition, signed images, HTTPS from lot-systems.com.
- Storage: NVS for device ID, token, Wi-Fi creds, last notification ID.

2. State machine

```
DEEP_SLEEP --timer/button--> WAKE --> WIFI_CONNECT --> SYNC
SYNC: GET /api/device/notifications?since=<id>
      POST /api/device/telemetry (weather, battery) every Nth wake
      POST /api/device/events (queued button events)
SYNC --new notification--> DISPLAY (screen on, buzz) --timeout 20s/Copy--> DEEP_SLEEP
```

Poll interval: 60–300 s (config from server). Button is a GPIO wake source, so Copy works even between polls (event queued in RTC memory, sent on next connect).

3. Display

Screen shows: message (max ~60 chars, 2–3 lines), time, battery. Terminal-grid style (monospace, no emoji, periods not symbols), per LOT-STYLE-GUIDE.

4. Sensors

BME688: temp/humidity/pressure every poll; gas/IAQ via Bosch BSEC when awake ≥ 3 s (optional). Values rounded and sent as one JSON object.

5. Camera

Off by default. Capture only on a long-press of the button; status LED forced on during capture; JPEG ≤ 100 KB uploaded to /api/device/events (type photo). No continuous streaming; no audio hardware.

6. Self-test (factory jig)

Reports PASS/FAIL JSON over USB/UART pogo: flash ID, PSRAM, Wi-Fi scan, display pattern, BME688 ID, camera ID, button, battery voltage, charge IC status.

7. Firmware document set (item 9)

Doc	Content	Format
FW-ARCH	this spec, expanded with task/memory maps	MD
FW-PINMAP	GPIO assignments per board rev	MD/CSV
FW-PROTOCOL	wire format; mirrors 04	MD
FW-BUILD	toolchain, build, flash, sign, OTA	MD
FW-RELEASE-NOTES	per version	MD
MANUAL	user manual, PDF	PDF

Repo layout proposed: hardware/firmware/ (ESP-IDF project) and hardware/tools/ (host software).

04 — LOT API CONNECTOR & HOST SOFTWARE

1. Existing surface (from src/server/routes/api.ts)

- GET /api/live-message — single global message string (no per-device targeting).
- POST /api/logs {text, event?, metadata?} — creates a Log row for the session user (session-cookie auth).
- GET /api/weather — user-city weather.

Devices cannot use session cookies, so a device-token layer is required.

2. New endpoints (proposed, not yet built)

Auth: Authorization: Bearer <device-token>; token is created in the user's settings, hashed in DB, revocable, scoped to one user.

Method	Path	Purpose		
POST	/api/device/pair	Exchange one-time pairing code for a device token		
GET	/api/device/notifications?since=ID	Pending notifications for this device (JSON array, ≤5)		
POST	/api/device/notifications/:id/ack	Mark displayed		
POST	/api/device/events	{type: "copy"}	"photo"	"boot", ts, payload}
POST	/api/device/telemetry	{tempC, humidity, pressure, gas, batteryPct, fw}		
GET	/api/device/config	Poll interval, timezone, brightness		

New Prisma models: Device (id, userId, tokenHash, name, lastSeen, fw), DeviceNotification (id, deviceId, text, createdAt, ackedAt, source).

3. "Copy" → Log tab

On Copy the server writes via the existing Log model:

```
Log.create({ userId, text: "Copy pressed on LOT Computer.", event: "device_copy",  
  metadata: { deviceId, ts, notificationId, text } })
```

Logs.tsx shows event: device_copy rows with the notification text. No client change is needed beyond a label for the event type. Rate limit: 10 events/min/device.

4. Pager-like notifications from the AI site

- A scheduled job (see src/server/scheduled-jobs.ts) asks the AI engine for a short nudge using the user's context/memory engine; output hard-limited to 60 chars, no emoji.
- Insert DeviceNotification; device picks it up at the next poll.
- Phase 2: WebSocket/MQTT push for instant delivery (costs battery; optional).

5. Host software (item 10)

hardware/tools/lot-device (Node/TypeScript CLI, matches repo stack):

- lot-device flash <port> — esptool wrapper
- lot-device provision — writes device ID + token + Wi-Fi to NVS
- lot-device test — runs self-test, prints report
- lot-device simulate — software device for P1 (exercises the API without hardware)
- lot-device logs — serial monitor

6. Security

HTTPS only (pin the lot-systems.com CA chain), per-device revocable tokens, no secrets in firmware images, signed OTA, camera upload size and rate caps.

05 — MECHANICAL, CHARGING, MANUFACTURING

1. Enclosure

- Two parts, 316L: **front bezel** (window for screen, camera hole, button) and **back plate** (flat, mirror-polished, no holes).
- 40 × 40 mm, corner radius ~3 mm, total thickness target 5.0 mm (see R1). Wall 0.5–0.6 mm.
- Join: laser-weld is permanent; use a recessed snap + adhesive gasket (serviceable) for EVT/DVT.
- Finish: back plate mirror polish ($R_a < 0.05\ \mu\text{m}$), front bezel brushed or polished (pick one).
- Camera: sapphire or glass lens window flush with bezel, LED beside it.
- Antenna: non-metal window or insulating split between the two steel parts, tuned in EVT.

2. Stack (thickness budget, mm)

Layer	Target
Back plate	0.5
Qi coil + ferrite	0.5
Battery (pouch)	2.0
Flex PCB	0.4
Display + glass	1.4 (side-by-side with battery)
Front bezel	0.5

Total with side-by-side layout $\approx 4.9\text{--}5.5$. Camera (≥ 2.5) must sit beside the battery, not above it.

3. Wireless charging

Metal blocks Qi. Options, ranked:

1. **Front-face charging**: coil behind the glass/display area, device placed screen-down on a pad. Keeps the polished back untouched; weakest for daily use.
2. **Non-metal inlay in the back plate** (ceramic/sapphire disc) over the coil: efficient, breaks the all-steel look.
3. **Thin slotted steel** ($\leq 0.3\ \text{mm}$) + ferrite: lossy, heats; test only.

Plan: EVT bench test of 316L 0.3/0.5/0.8 mm over a Qi coil, measure received power and temperature. Charge target $\geq 200\ \text{mA}$ at 5 V in. Cell protection and 4.2 V charge termination in hardware.

4. PCBWay

- PCB: 4-layer, 0.6–0.8 mm, ENIG, impedance-controlled for the camera/display lines; consider flex-rigid for height.
- Turnkey PCBA with the BOM in 02. Gerber + BOM + CPL; ask for DFM review before the first order.
- Steel: PCBWay CNC machining service or Xometry; polish quote separate.

5. Compliance (if sold or distributed)

FCC Part 15 (module-pre-certified helps but metal case + antenna change requires re-test), CE-RED, WPC Qi, UN38.3 battery, RoHS/REACH. Budget and time are in 01 §4.

6. Privacy & safety

Camera LED on during capture, no always-on capture, no microphone. Manual documents these. Battery protection circuit mandatory.